

# Computer Vision Project – Exercise 04

## Group Report

Group Submission

July 6, 2026

## 1 Scope of the Group Submission

This report covers the group part of Exercise 04:

- Exercise 4.1: Face detection, tracking, and alignment
- Exercise 4.2: Face identification and open-set recognition
- Exercise 4.3: Face clustering and re-identification
- Exercise 4.4: DIR evaluation

Exercise 4.5 is treated as the individual challenge and is therefore not included in this group report.

## 2 Face Detection, Tracking, and Alignment

The face preprocessing pipeline consists of face detection, tracking, and alignment. MTCNN is used for face detection. The detected face is then aligned to a fixed resolution of  $224 \times 224$  pixels before feature extraction.

For video processing, template matching was implemented in the `track_face` method. The first frame is initialized using MTCNN. For later frames, a local search window around the previous face position is used. This avoids running MTCNN on every frame and makes the system closer to the intended video tracking pipeline.

### 2.1 Tracking Drift and Fix

During testing, a template drift problem was observed. The tracker sometimes slowly moved away from the actual face while still producing a high matching score. This especially affected the Alan\_Ball sequence, where recognition was initially very poor.

To reduce this problem, periodic MTCNN re-detection was added. The idea is to keep template tracking for efficiency, but re-anchor the tracker after a fixed number of frames.

The following re-detection intervals were tested:

| Setting                             | Description                                        | Alan_Ball Result |
|-------------------------------------|----------------------------------------------------|------------------|
| <code>redetect_interval</code> = 15 | Less frequent MTCNN re-detection                   | 277/432          |
| <code>redetect_interval</code> = 5  | Practical default balancing runtime and robustness | 385/432          |
| <code>redetect_interval</code> = 1  | MTCNN on every frame, highest accuracy but slowest | 426/432          |

Table 1: Effect of periodic re-detection on the Alan\_Ball sequence.

The final submitted group code uses:

```
redetect_interval = 5
```

This was selected as the practical default because it still uses template tracking and avoids the high computational cost of running MTCNN on every frame. The `redetect_interval = 1` result is reported as an additional accuracy-focused experiment.

### 3 Face Identification

Face identification was implemented using FaceNet embeddings and a k-nearest neighbor classifier. The implementation does not use sklearn for the group part.

The implemented components are:

- FaceNet embedding extraction from aligned faces
- Gallery construction using `partial_fit`
- k-nearest neighbor prediction using Euclidean distances
- majority voting over the  $k$  nearest neighbors
- posterior probability computed as  $k_i/k$
- distance to the predicted class computed using the nearest neighbor from the predicted class
- color and grayscale embeddings stored for each training face
- prediction using both color and grayscale query embeddings

The final parameters are:

| Parameter                      | Value |
|--------------------------------|-------|
| <code>num_neighbours</code>    | 3     |
| <code>max_distance</code>      | 1.02  |
| <code>min_prob</code>          | 0.34  |
| <code>redetect_interval</code> | 5     |

Table 2: Final parameters used in the group source code.

### 4 Open-Set Recognition

The open-set decision rule predicts a face as unknown if the distance to the predicted class is too high or the posterior probability is too low:

```
if distance_to_prediction > max_distance or posterior < min_prob:  
    prediction = "unknown"
```

The unknown identities used in testing were:

- Al\_Pacino
- Maria\_Shriver

A threshold sweep was performed after fixing the tracker. For `redetect_interval = 5`, the best balanced threshold was:

| Threshold | Known Accuracy     | Unknown Rejection | Balanced Score |
|-----------|--------------------|-------------------|----------------|
| 1.02      | 1040/1080 = 96.30% | 154/154 = 100%    | 98.15%         |

Table 3: Best threshold from the post-tracking-fix threshold sweep.

## 5 Identification Results

The final practical code setting uses `redetect_interval = 5`. Selected final-code results are:

| Identity      | Result                      |
|---------------|-----------------------------|
| Alan_Ball     | 385/432 = 89.12%            |
| Al_Pacino     | 48/49 rejected as unknown   |
| Maria_Shriver | 105/105 rejected as unknown |

Table 4: Selected final-code identification results with `redetect_interval = 5`.

An additional high-accuracy experiment was performed with `redetect_interval = 1`. This setting is slower because it effectively uses MTCNN every frame, but it shows the upper accuracy that can be reached when tracking drift is removed.

| Identity          | Result                      |
|-------------------|-----------------------------|
| Alan_Ball         | 426/432 = 98.61%            |
| Manuel_Pellegrini | 229/229 = 100%              |
| Marina_Silva      | 173/173 = 100%              |
| Nancy_Sinatra     | 67/73 = 91.78%              |
| Peter_Gilmour     | 173/173 = 100%              |
| Al_Pacino         | 49/49 rejected as unknown   |
| Maria_Shriver     | 105/105 rejected as unknown |

Table 5: Accuracy-focused experiment with `redetect_interval = 1`.

Overall result for the `redetect_interval = 1` experiment:

- Known-person accuracy: 1068/1080 = 98.89%
- Unknown rejection: 154/154 = 100%

## 6 Face Clustering and Re-Identification

The `FaceClustering` class implements k-means clustering in FaceNet embedding space.

Implemented components:

- store unlabeled FaceNet embeddings
- estimate cluster centers with k-means
- store cluster assignments
- predict the closest cluster for a query face
- return the distance distribution to all cluster centers

Clustering was tested with two persons:

- Alan\_Ball
- Manuel\_Pellegrini

Saved clustering evidence includes:

- `results/models/clustering_gallery_final.pkl`
- `results/logs/cluster_predictions.csv`
- `results/screenshots_cluster/`

## 7 DIR Evaluation

The DIR evaluation was completed using the provided evaluation embeddings. The final metrics are:

| Metric                                  | Value     |
|-----------------------------------------|-----------|
| DIR@FAR=1%                              | 0.707091  |
| Threshold@1%                            | -0.497051 |
| DIR@FAR=10%                             | 0.883659  |
| Threshold@10%                           | -0.646075 |
| Minimum FAR with $\text{DIR} \geq 90\%$ | 0.140605  |
| Threshold there                         | -0.671862 |

Table 6: Final DIR evaluation results.

The DIR curve was saved as:

results/plots/dir\_curve.png

## 8 Conclusion

The group part implements the full face recognition pipeline: detection, tracking, alignment, supervised face identification, open-set rejection, clustering-based re-identification, and DIR evaluation.

The most important improvement was fixing template-tracking drift through periodic MTCNN re-detection. The final source code uses `redetect_interval = 5` as a practical runtime-aware setting, while the report also documents the higher-accuracy `redetect_interval = 1` experiment.